APMA 4302: Methods in Computational Science

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Lecture 20

Multi-physics problems in firedrake

Outline

- Model problem: thermal convection
- Basic physics:
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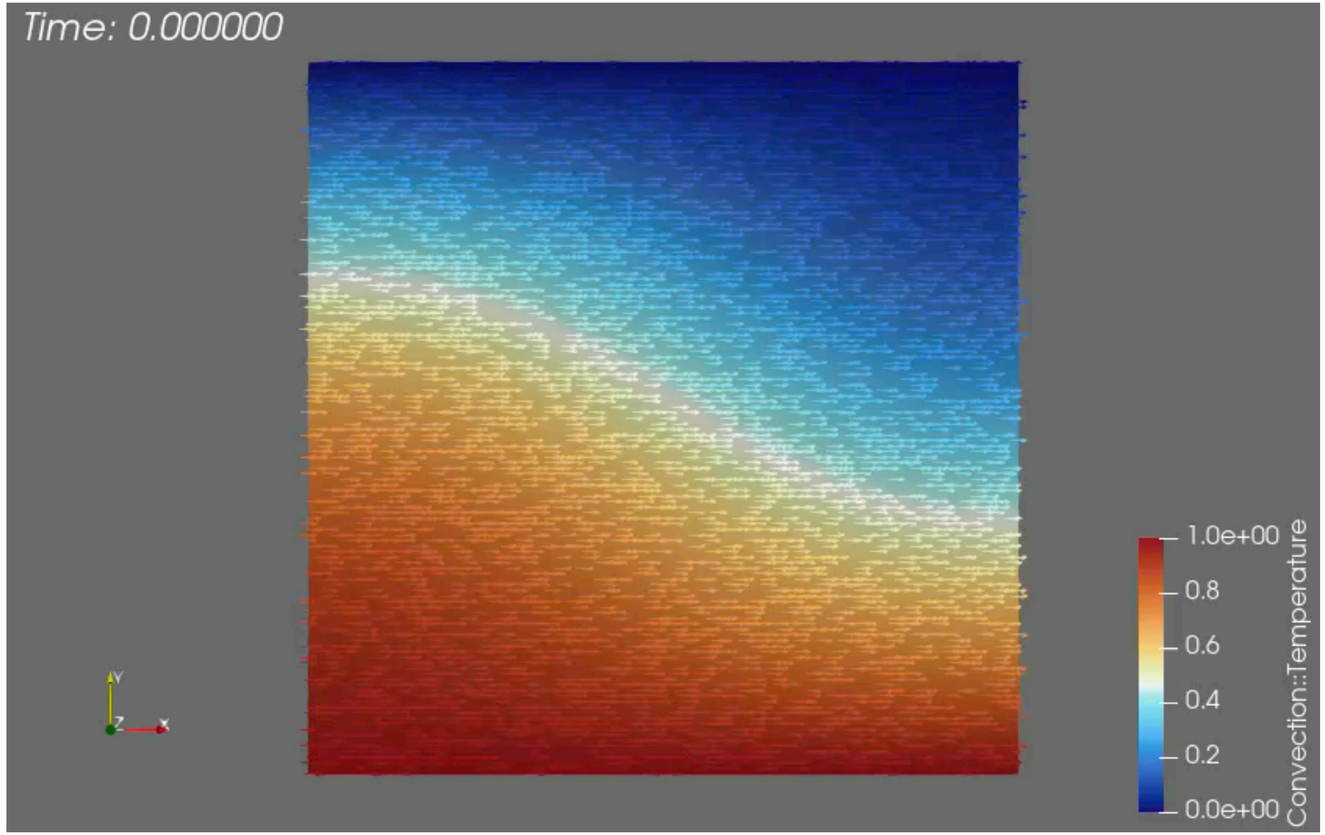
Infinite Pr # Thermal convection



Advection-Diffusion Eq.

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \frac{1}{Ra} \nabla^2 T$$

$$-\nabla \cdot \eta \left[(\nabla \mathbf{v}) + (\nabla \mathbf{v})^T \right] + \nabla P = -T\hat{\mathbf{g}}$$

$$\nabla \cdot \mathbf{v} = 0$$

where $Ra = \dfrac{\rho_0 \alpha \Delta T h^3}{\eta \kappa}$ is the Raleigh #

and $\eta = \eta(T, \mathbf{v})$ is the viscosity

Incompressible Stokes flow

Problem #1: Infinite Prandtl number thermal convection

Infinite Pr # Thermal convection

Streamfunction-vorticity formulation

Let $\mathbf{v} = \nabla \times (\psi \hat{\mathbf{k}}),\ \nabla \times \mathbf{v} = \omega \hat{\mathbf{k}}$

where ψ is the stream function

and ω is the voriticity

$$\frac{\partial T}{\partial t} + \nabla \times (\psi \hat{\mathbf{k}}) \cdot \nabla T = \frac{1}{Ra} \nabla^2 T$$

$$-\nabla^2 \omega = \frac{\partial T}{\partial x}$$

$$-\nabla^2 \psi = \omega$$

where $Ra = \dfrac{\rho_0 \alpha \Delta T h^3}{\eta \kappa}$ is the Raleigh #

With BC's and IC's

Blankenbach Benchmarks

Geophys. J. Int. (1989) **98**, 23–38

A benchmark comparison for mantle convection codes

B. Blankenbach†, F. Busse‡, U. Christensen§, L. Cserepes¶,
D. Gunkel††, U. Hansen††, H. Harder§, G. Jarvis#,
M. Koch§§, G. Marquart¶¶, D. Moore*, P. Olson**
H. Schmeling††† and T. Schnaubelt‡

Metrics:

Benchmark 1a, iso-viscous steady state square box $Ra = 10^4, 10^5, 10^6$

(i) The Nusselt number

$$\mathrm{Nu} = -h \frac{\int_0^l \partial_z T(x, z=h)\, dx}{\int_0^l T(x, z=0)\, dx}$$

(ii) The (non-dimensional) rms velocity

$$v_{\mathrm{rms}} = \frac{h}{\kappa}\left\{\frac{1}{hl}\int_0^l \int_0^h (u^2 + w^2)\, dz\, dx\right\}^{1/2}.$$

(iii) Non-dimensional temperature gradients at the corners of the cell:

$$q = \frac{-h}{\Delta T}\left(\frac{\partial T}{\partial z}\right),$$

with

q_1 at $x = 0,\ z = h$; q_2 at $x = l,\ z = h$;

q_3 at $x = l,\ z = 0$; q_4 at $x = 0,\ z = 0$.

And more:

TerraFERMA results (fenics+Petsc)

Table 2.1: Results from 2D, isoviscous square convection benchmark cases [1].

Case 1a: Ra $= 10^4$

	Nu	v_{rms}	q_1	q_2	T_e	z_e
16×16	4.897	42.884	8.062	0.588	0.423	0.233
32×32	4.887	42.865	8.060	0.589	0.422	0.226
64×64	4.885	42.865	8.059	0.589	0.422	0.226
128×128	4.885	42.865	8.059	0.589	0.422	0.225
256×256	4.884	42.865	8.059	0.589	0.422	0.225
Benchmark	4.884	42.865	8.059	0.589	0.422	0.225

Case 1b: Ra $= 10^5$

	Nu	v_{rms}	q_1	q_2	T_e	z_e
16×16	10.570	193.493	19.105	0.722	0.431	0.111
32×32	10.539	193.222	19.081	0.722	0.428	0.114
64×64	10.535	193.215	19.080	0.723	0.428	0.111
128×128	10.534	193.215	19.080	0.723	0.428	0.112
256×256	10.534	193.214	19.079	0.723	0.428	0.112
Benchmark	10.534	193.214	19.079	0.723	0.428	0.112

Case 1c: Ra $= 10^6$

	Nu	v_{rms}	q_1	q_2	T_e	z_e
16×16	22.107	836.687	46.187	0.896	0.434	0.061
32×32	21.982	834.024	46.008	0.877	0.432	0.059
64×64	21.971	833.990	45.972	0.877	0.432	0.058
128×128	21.972	833.989	45.967	0.877	0.432	0.058
Benchmark	21.972	833.990	45.964	0.877	0.432	0.058